Failure Analysis Report

FAIL-0239 — Connectivity Failure

1. Failure Summary

Failure ID	FAIL-0239
Failure Type	Connectivity Failure
Severity	Trivial
Component	Voice Pipeline
Application	APP-002 — MobileCommunication
Build Version	6.3.9
Timestamp	2023-11-21 15:44:33
Environment	ENV-006 — Cloud Test Env
OS Version	Ubuntu 22.04
Network	Ethernet
Resolution	Open

2. Test Case Reference

Test Case ID	TC-0649
Title	Verify end-to-end flow of session token refresh from cart selection to pairing c...
Execution ID	EXEC-01122
Test Run	RUN-003 — Regression Cycle 3 — Build 4.3.2
Requirement	REQ-0399 — Compatibility — Push Notification Delivery

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete Bluetooth reconnection within the defined SLA.
Observed Behaviour	The system exhibited connectivity failure when the input data contains Unicode special characters.

4. Error Details

Error Message:

```
BluetoothException: device DEVICE-157 disconnected during pairing handshake.
```

Stack Trace / Log Excerpt:

```
com.samsung.wallet.FirmwareManager.execute(line 699)
```

Log Reference	LOG-39091

5. Executed Steps at Time of Failure

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1	Observe the status indicator for up to 16 seconds.	The performance counter logs a value below the SLA threshold.
2	Initiate push notification delivery by pressing the 'Sync' button...	No crash or ANR is observed.
3	Verify that the UI displays 'Connection successful'.	The device reconnects automatically within 10 seconds.
4	Observe the status indicator for up to 23 seconds.	The performance counter logs a value below the SLA threshold.

6. Root Cause Analysis

Cache eviction policy evicts frequently-used configuration entries under memory pressure.

7. Linked Defect

Defect ID	BUG-0469
Title	Connectivity Failure in Voice Pipeline: BluetoothException: device DEVICE-157 di...
Severity	Trivial
Priority	Low
Status	OPEN
Root Cause	Cache eviction policy evicts frequently-used configuration entries under memory ...
Resolution	—
Assigned To	ENG-073

8. Historical Similar Failures

HIST-0238	FAIL-0239	Connectivity Failure	6.1.4	2022-05-11	11	Resolved
HIST-0728	FAIL-0729	Connectivity Failure	5.4.0	2021-10-27	1	Resolved
HIST-0807	FAIL-0808	Connectivity Failure	3.2.0	2023-01-12	13	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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